

NITISHKUMAR KHAVEKAR

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PROFESSIONAL SUMMARY

AI/ML Engineer Fresher with hands-on experience building end-to-end machine learning pipelines (Scikit-learn, Python) and production-ready Generative AI systems including RAG chatbots (LangChain, Gemini API, FAISS) with semantic search and vector embedding capabilities. Proficient in Python, Streamlit dashboarding, and automating data pipelines to deliver measurable business and educational analytics insights.

INTERNSHIP EXPERIENCE

DATA ANALYST INTERN

KAMJRITZTEX IT SOLUTIONS | Jun 2025 – Dec 2025

- Analyzed data from 3 critical business modules to identify workflow bottlenecks and performance trends, driving process improvements that increased operational efficiency by 30%.
- Developed and optimized SQL queries for data extraction, validation, and analysis, reducing data retrieval and reporting time by 25%.
- Developed automated data pipelines and analytics dashboards to extract, transform, and monitor business KPIs — providing the data infrastructure foundation for predictive modeling and ML feature engineering workflows.

PROJECTS

RAG Based AI Chatbot | Python | LangChain | Streamlit | FAISS | Scikit-Learn

May 2026 - Jun 2026

- Architected a Python-based document processing pipeline extracting text from multi-format files, recursively chunking data, and generating 768-dimensional dense vector embeddings using Google Gemini API to enable semantic search.
- Developed a local FAISS vector store indexing 10,000+ text chunks for semantic retrieval, serving context to a gemini-2.5-flash LLM to generate accurate responses with page-level citations.
- Deployed a custom Streamlit web interface in Python featuring side-by-side RAG controls and general chat, reducing manual information lookup time by 40% for administrators querying internal knowledge bases.

Student Performance Predictor | Python | Streamlit | Scikit-Learn | Data Pipelines

Mar 2026 – Apr 2026

- Developed an end-to-end machine learning system using Python and Scikit-learn to analyze 1,000+ student academic profiles, achieving 91% R^2 accuracy with a Decision Tree Regressor to predict and identify at-risk students.
- Implemented a modular data preprocessing pipeline utilizing custom ColumnTransformers for feature scaling and one-hot encoding, facilitating real-time predictions and bulk CSV batch inference to identify underperforming students.
- Deployed an interactive Streamlit dashboard featuring multi-page navigation, EDA visualizations of demographic KPIs, and automated model-retraining triggers, reducing administrative reporting overhead by 50% through self-service analytics.

Hospital Appointment No-Show Prediction & Analysis | SQL | Power BI

Feb 2026 - Mar 2026

- Built an end-to-end data analytics pipeline on 110,000+ hospital appointment records using SQL for data extraction and Python for data preprocessing, feature engineering, and exploratory data analysis to identify appointment no-show patterns.
- Performed feature engineering and statistical analysis to identify key drivers of appointment no-shows and high-risk patient groups.
- Developed interactive Power BI dashboards to visualize patient risk segments, no-show trends, and operational KPIs, enabling healthcare administrators to design data-driven interventions aimed at reducing missed appointments.

SKILLS

Programming Languages: Python | SQL | Bash

Machine Learning: Scikit-learn | Supervised Learning | Unsupervised Learning | Regression | Classification | Clustering | Feature Engineering | Model Evaluation | Cross-Validation | Hyperparameter Tuning

Deep Learning: PyTorch | Neural Networks | CNN | RNN | LSTM | Transformers

Generative AI & LLMs: Large Language Models (LLMs) | Prompt Engineering | Retrieval-Augmented Generation (RAG) | AI Chatbots

Natural Language Processing: Text Preprocessing | Tokenization | Text Classification | Sentiment Analysis

Data Science & Visualization: Pandas | NumPy | Matplotlib | Seaborn

Cloud Platforms: AWS | Azure

EDUCATION

SINHGAD COLLEGE OF ENGINEERING

Bachelor of Technology in Computer Science
CGPA: 9.1

Solapur, MH
Feb 2021 - Jun 2025

CERTIFICATES

Advance certificate in Data Science – upGrad | 2026

E-commerce Customer Review Analytics Mentorship – Snapdeal & upGrad | 2026